

Anonymous Equal-Amplitude Composite-Wave Model: Basic Axiom System v4

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1. First Principles

Axiom 0: Component Anonymity

Classification: Axiom

Basic components are not assigned individual names, privileged axes, privileged signs, or privileged types.

The index **n** is a convenient label and is not an intrinsic name of the component.

Forbidden:

making only one specific component the negative-sign axis
making only one specific component the time axis
making only one specific component real-valued
making only one specific component complex-valued
giving an intrinsic name to only one specific component

Axiom 0+: Formulation Anonymity

Classification: Axiom

At the first-principle layer, theory names, formulation names, and readout-logic names are not assigned intrinsic names.

Equations, projections, or readout rules at the first-principle layer must not be changed on the basis of physical names, object names, interaction names, or existing theory classifications.

Forbidden:

adopting a circular-motion equation because it is called gravity
adopting a hyperbolic equation because it is called Coulomb force
erasing signs because it is called mass
preserving signs because it is called charge
moving a component to a time axis because it is called energy
using a different closure equation because the target is a two-body system
using a different closure equation because the target is a three-body system
using a special readout rule because the target is an ABC system
choosing a right-hand-side representation to obtain a desired physical name

Allowed:

different readout operations applied to the same closure condition
different symmetry breaking applied to the same closure condition
different gauge stability applied to the same closure condition
different sign preservation applied to the same closure condition
different sign erasure applied to the same closure condition
different phase-branch selection applied to the same closure condition
different closure target sets applied to the same closure condition

Order:

place the same all-positive-sign zero closure
define the readout operation first
define symmetry breaking first
define sign preservation or sign erasure first
confirm gauge stability
assign working names after readout

Target set:

$$Q(P)=\sum_n p_n^2$$

$$Q(A)=0$$

$$Q(A+B)=0$$

$$Q(A+B+C)=0$$

Differences in target set are allowed.

Changing the definition of Q is not allowed.

Axiom 0.5: Dimension-Wise Central Projection Readout

Classification: Auxiliary geometric axiom

No curvature correction is introduced for readout along a single axis.

Curvature-correction candidates are introduced only when two or more independent directions form an area, volume, closed path, or transport mismatch.

Axiom 1: All-Positive-Sign Zero Closure

Classification: Axiom

$$\sum_{n=1}^N x_n^2=0$$

Expanded form:

$$x_1^2+x_2^2+\cdots+x_N^2=0$$

Coefficient signs:

all terms +

Not adopted:

$$\sum_n |x_n|^2=0$$

Term used for closure:

$$x_n^2$$

Term not used for closure:

$$x_n\bar{x}_n$$

Applicable states:

stationary closed state

stable closed state

Quasi-stationary states:

immediately after external influence

closure recovery process

metastable state

Temporary nonzero closure residual in a quasi-stationary state:

$$\sum_n x_n^2 \neq 0$$

After stationarization:

$$\sum_n x_n^2 = 0$$

Axiom 2: Nontrivial Existence

Classification: Axiom

$$\exists n, \text{quad } x_n \neq 0$$

Axiom 3: Projection-Axis Degeneration and Imaginary-Direction Readout

Classification: Axiom

When an intrinsic coordinate system is constructed by a projection method, the projection-axis direction is not distinguishable as a direction for the residents of that intrinsic coordinate system.

However, if a value in the projection-axis direction remains inside the closure condition, that value is not erased.

When the compensating quantity in the projection-axis direction is read as z , it is treated inside the intrinsic coordinate system as

$$iz$$

This component is unobservable as a first-order direction.

When squared inside the closure expression,

$$(iz)^2 = -z^2$$

it appears with a reversed sign.

Therefore, an unobservable compensating quantity in the projection-axis direction appears inside the intrinsic coordinate system as a sign-reversed compensating quantity through squared readout.

2. Consequences from the First Principles

Consequence 1: Insufficiency of Real Positive-Definite Components

Classification: Derived consequence

$$x_n \in \mathbb{R}$$

$$\Rightarrow$$

$$x_n^2 \geq 0$$

$$\sum_n x_n^2 = 0$$

$$\Rightarrow$$

$$x_n = 0 \text{ quad for all } n$$

Consequence 2: Internal Generation of Sign Reversal

Classification: Derived consequence

$$x_1 = A, \text{quad } x_2 = iA$$

$$x_1^2 + x_2^2 = A^2 + (iA)^2 = A^2 - A^2 = 0$$

$$i^2 = -1$$

Consequence 3: Necessity of Phase Algebra

Classification: Derived consequence

$$x_n = r_n e^{i\phi_n}$$

$$x_n^2 = r_n^2 e^{i2\phi_n}$$

Closure condition:

$$\sum_{n=1}^N r_n^2 e^{i2\phi_n} = 0$$

Equal-amplitude condition:

$$r_n = r$$

$$\sum_{n=1}^N e^{i2\phi_n} = 0$$

Consequence 4: Type Anonymity

Classification: Derived consequence

$$x_n \in \mathbb{C} \quad \forall n$$

or

all components are assigned the same phase-algebra type

Consequence 5: Ninety-Degree Phase Difference

Classification: Derived consequence

$$A^2 + (iA)^2 = 0$$

3. Anonymous Equal-Amplitude Composite Wave

Axiom 4: Anonymous Equal-Amplitude Composite Wave

Classification: Axiom

$$\Psi_N = \frac{A_0}{N} \sum_{k=1}^N e^{i\theta_k}$$

$$|\Psi_k| = \frac{A_0}{N}$$

When all phases are aligned:

$$\max |\Psi_N| = A_0$$

Axiom 5: Unreadability of Individual Phase

Classification: Axiom

$$\theta_k \rightarrow \theta_k + \alpha$$

Axiom 6: Phase-Difference Readout

Classification: Axiom

$$\Delta_{ij} = \theta_i - \theta_j$$

Number of readable pair channels:

$$\binom{N}{2} = \frac{N(N-1)}{2}$$

Axiom 7: System Identification

Classification: Axiom

$$(N, A_0, \{\Delta_{ij}\})$$

Axiom 8: Composite Phase Projection

Classification: Axiom

$$S_N = \sum_k e^{i\theta_k}$$

$$\Psi_N = \frac{A_0}{N} S_N$$

$$\Theta_N = \arg \Psi_N$$

Axiom 9: No Requirement of Inversion Symmetry

Classification: Axiom

Inversion symmetry is not required.

4. External Readout

Axiom 10: External-Axis Response

Classification: Axiom candidate

$$M_\alpha = e^{i\alpha}$$

$$Y(\alpha) = \frac{A_0}{N} \sum_{k=1}^N e^{i(\theta_k - \alpha)} \\ = e^{-i\alpha} \Psi_N$$

Continuous response:

$$S_N(\alpha) = \operatorname{Re} \left[\frac{1}{N} \sum_{k=1}^N e^{i(\theta_k - \alpha)} \right]$$

Binary response:

$$S_N(\alpha) = \operatorname{sgn} \left[\operatorname{Re} \left[\frac{1}{N} \sum_{k=1}^N e^{i(\theta_k - \alpha)} \right] \right]$$

Axiom 11: External Readout Phase

Classification: Axiom candidate

$$\Delta \phi_\mu = \phi_\mu(\Psi) - \phi_\mu(M)$$

$$\mu = x, y, z, t$$

Axiom 12: Position-Phase Readout

Classification: Axiom candidate

x, y, z

Axiom 13: Time-Phase Readout

Classification: Axiom candidate

$\phi_t \equiv \omega t_{\text{read}} \pmod{2\pi}$

Axiom 14: Covering Phase

Classification: Axiom candidate

2π -periodic readout

4π -periodic readout

winding-number-difference readout

5. Observational Phase Discreteness

Axiom 15: Observational Phase Discreteness

Classification: Axiom candidate

$\sum_{\text{cycle}} \Delta \phi = 2\pi m, \text{quad } m \in \mathbb{Z}$

$W = \frac{1}{2\pi} \oint d\phi, \text{quad } W \in \mathbb{Z}$

Readout-cell form:

$\phi_{\text{obs}} = m \epsilon_{\phi}, \text{quad } m \in \mathbb{Z}$

Phase-area candidate:

$Q_{\phi} = \sum_{i < j} \sin^2 \frac{\Delta \phi_{ij}}{2}$

6. Waveform and Localization

Working Axiom A: State Waveform

Classification: Working hypothesis

$A(\theta) = \sum_n A_n \cos(n\theta + \phi_n)$

$Z(\theta) = \sum_n r_n e^{i(n\theta + \phi_n)}$

Definition A1: Normalized Equal-Amplitude Odd-Harmonic Localized Wave

Classification: Definition

$N_h = 2K - 1$

$S_{N_h}(u) = \frac{1}{K} \sum_{m=0}^{K-1} \cos((2m+1)u)$

$Z_{N_h}(u) = \frac{1}{K} \sum_{m=0}^{K-1} e^{i(2m+1)u}$

$\Psi_{N_h}(u) = A S_{N_h}(u)$

$$\Psi_{N_h}(u) = A \frac{Z_{N_h}(u)}{\frac{A}{K}}$$

Theorem A1: Representative-Amplitude Preservation

Classification: Derived theorem

$$S_{N_h}(0) = 1$$

$$\Psi_{N_h}(0) = A$$

Theorem A2: Highest Odd-Harmonic Order and Localization Width

Classification: Derived theorem

$$K = \frac{N_h + 1}{2}$$

$$\lambda_{N_h} = \frac{\lambda_0}{N_h}$$

$$\Delta x_{N_h} \propto \frac{\lambda_0}{N_h}$$

$$N_{h, \text{req}} \sim \frac{\lambda_0}{\Delta x}$$

$$N_{h, \text{req}}^{(t)} \sim \frac{T_0}{\Delta t}$$

Consequence A1: Identical Construction of Spatial and Temporal Localization

Classification: Derived consequence

$$\Psi(\chi, \tau) = A S_{N_h, \chi}(\chi - \chi_0) S_{N_h, \tau}(\tau - \tau_0) e^{i\phi}$$

Working Axiom B: Localization

Classification: Working hypothesis

$$S_{N_h}(u) = \frac{1}{K} \sum_{m=0}^{K-1} \cos((2m+1)u)$$

$$Z_{N_h}(u) = \frac{1}{K} \sum_{m=0}^{K-1} e^{i(2m+1)u}$$

Working Axiom C: Coupling Entrance

Classification: Working hypothesis

The entrance to interaction is low-order base-phase alignment.

Working Axiom D: Observation Map

Classification: Working hypothesis

$$s_\alpha = \text{Loc} \left[\int A(\theta) M_\alpha(\theta) d\theta \right]$$

Working Axiom E: Squared Correlation Strength

Classification: Working hypothesis

$$P(\alpha) = \frac{|C_\alpha|^2}{\sum_\beta |C_\beta|^2}$$

Working Axiom F: Shared Low-Order Base Synchronization

Classification: Working hypothesis

$$\theta_A - \theta_B = \Delta$$

Working Axiom G: Synchronized Demodulation Failure

Classification: Working hypothesis

$$V(t) = \left| \sum_n w_n e^{i\epsilon_n(t)} \right|$$

$$V(t) = \left| \sum_n w_n e^{-\frac{1}{2}\sigma_n^2(t)} \right|$$

Working Axiom H: Hierarchy

Classification: Working hypothesis

low-order base mode

+ high-order localized mode

7. Readout, Memory, and Complex Representation

I/Q Readout

Classification: Theoretical observation

$$z = a + ib$$

$$a = r \cos \theta, \quad b = r \sin \theta$$

$$u = r \cos \theta + r \sin \theta$$

$$= \sqrt{2} r \cos \left(\theta - \frac{\pi}{4} \right)$$

Harmonic-Information Readout

Classification: Theoretical observation

$$Z(\theta) = \sum_n r_n e^{i(n\theta + \phi_n)}$$

$$P(Z) = \operatorname{Re}(Z) + \operatorname{Im}(Z)$$

$$P(Z) = \sum_n \sqrt{2} r_n$$

$$\cos \left(n\theta + \phi_n - \frac{\pi}{4} \right)$$

Preservation Side and Readout Side

Classification: Theoretical observation

$$\bar{z} = a^2 + b^2$$

$$z^2 = a^2 - b^2 + 2iab$$

Square Registry

Classification: Working hypothesis

$$\mathcal{A}_0 = \sum_k A_k^2$$

$$a_0 = \sum_k |A_k|^2 = \sum_k A_k \bar{A}_k$$

8. Statistical Classification

Pair Phase Difference

Classification: Definition

$$\Delta_{ij} = \theta_i - \theta_j$$

$$\binom{N}{2} = \frac{N(N-1)}{2}$$

Pair composite:

$$\begin{aligned} \psi_i + \psi_j \\ = 2A \cos \frac{\Delta_{ij}}{2} e^{i(\theta_i + \theta_j)/2} \end{aligned}$$

Pair interference strength:

$$I_{ij} = 4A^2 \cos^2 \frac{\Delta_{ij}}{2}$$

Overlap Degree

Classification: Definition

$$R_N = \left| \frac{A_0}{N} \sum_{k=1}^N e^{i\theta_k} \right|$$

$$\begin{aligned} \rho_N &= \frac{R_N}{A_0} \\ &= \frac{1}{N} \left| \sum_{k=1}^N e^{i\theta_k} \right| \end{aligned}$$

$$0 \leq \rho_N \leq 1$$

Fermionic Degree

Classification: Definition

$$\begin{aligned} F_N &= \\ &= \frac{2}{N(N-1)} \\ &= \sum_{i < j} \sin^2 \frac{\Delta_{ij}}{2} \end{aligned}$$

$$B_N = 1 - F_N$$

$$\begin{aligned} B_N &= \\ &= \frac{2}{N(N-1)} \\ &= \sum_{i < j} \cos^2 \frac{\Delta_{ij}}{2} \end{aligned}$$

$$F_N = \frac{N}{2(N-1)} (1 - \rho_N^2)$$